Soumya Sharma

+91-7007913491 | btech15109.24@bitmesra.ac.in | linkedin.com/in/soumyasharma427 | github.com/Swizknife

Education

Birla Institute of Technology, Mesra (Patna Campus) — Patna, India
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 7.2/10 — *2024 – May 2028*

Children College, Azamgarh — Azamgarh, India
Class XII (ISC): 93% | *Class X (CISCE): 92.4%*

Experience

NS Apps Innovation — Sep 2026 – Nov 2026
AI/ML & React Developer Intern — *India*

- Built a **hybrid OKF + RAG** chatbot that answers Bihar MSME policy questions with citations. A Google **Open Knowledge Format (OKF)** knowledge graph (249 concepts, 311 edges) answers exact-fact queries in **0.02–0.13 s vs. 20–110 s** on the LLM path, while open questions flow through **BGE-M3** dense + sparse retrieval in **Qdrant**, rank fusion and a **bge-reranker-v2-m3** cross-encoder, gated by a coverage check that declines low-evidence answers. Served via **FastAPI** + React; 109 figures source-verified, 51 tests
- Engineered the backend of **Yoda**, an AI desk robot, on an **NVIDIA DGX Spark**: an asyncio **FastAPI** server ingests HTTPS frames into a drop-oldest queue and holds an authenticated **WebSocket** for commands and 16 kHz audio. A PyTorch-free **ONNX Runtime** vision chain (RTMO-s pose → YuNet → SFace → FER+) runs at **60–70 ms/frame, 5 FPS with 0 dropped frames** with keypoint-based wave detection and open-set face ID; a bilingual "Hey Yoda" voice loop chains Silero VAD, int8 **faster-whisper** (1.4 s), **Qwen3-VL-8B** on the live frame and Piper TTS. A Pydantic allow-list gates LLM commands to hardware; 167 pytest tests

Undergraduate Research Intern – Nanopore Biosensor Signal Processing — Aug 2026 – Present
Under Dr. Naren Das, Dept. of ECE, BIT Mesra — *Patna, India*

- Built a 17-module Python pipeline (NumPy, SciPy, scikit-learn); curated a **440-recording (3.1 GB)** dataset
- Event classifier reached **0.84 balanced accuracy** on an independent test day vs. 0.55 for a cross-session baseline; validated event fitting on 3,150 synthetic events (**92–100%** shape accuracy)

Projects

Amazon ML Challenge 2026 – Entity Resolution | *Python, LightGBM, PyTorch, Transformers* — GitHub

- Led a 3-member team matching 1.7M businesses against ∼**10M noisy records**; multi-key blocking + a two-stage **LightGBM** cascade (73 features) cut candidates from ∼300 to **5.8 per entity** at 0.979 recall
- Fine-tuned a multilingual **mmBERT-small (140M)** cross-encoder on Kaggle GPUs to re-score uncertain pairs (AUC 0.955 vs. 0.931 for LightGBM); final **public-leaderboard F0.5 of 0.968** among ∼90,000 registrants

Twigg – LLM Copilot (Browser Extension) | *JavaScript, Manifest V3, Gemini API, Ollama* — GitHub

- Built a **Manifest V3** extension: autofill engine, JSON-LD parsing, **Gemini API/Ollama**, **LLM guardrails**

MindCheck – Adaptive Mental-Health Screening Platform | *FastAPI, Next.js, NumPy, SQLAlchemy* — 2026

- Built the FastAPI + Next.js platform behind the publication: IRT adaptive screening; **92 backend tests**

Technical Skills

Languages & Frameworks: Python, C/C++, TypeScript, JavaScript, SQL, React
ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, ONNX Runtime, OpenCV, Pandas, Polars
LLM / GenAI: RAG, OKF knowledge graphs, Qdrant (vector DB), embeddings, rerankers, Ollama, Whisper, VLMs
Cloud / Backend: GCP Cloud Run, Docker, GitHub Actions CI/CD, FastAPI, WebSockets, React, Git, Linux

Applied Patent and Publications

- **Hybrid Psychometric-Machine Learning Platform for Multi-Modal Mental Health Screening of Engineering Students**: IRT-based adaptive mental-health screening

Achievements

- **Finalist**, TantraFiesta 2025, IIIT Nagpur
- **Finalist**, Coinbase-sponsored Web3 Hackathon, Innovate-a-thon 2025, BIT Mesra
- Cleared Round 1, **Adobe University Hackathon 2026**
- **Basketball Gold Medalist**: 2× winner at inter-branch and inter-college level

Leadership & Extracurriculars

- **Head of Publicity, TECHNIKA 26-27**: led flash-mob, social-media & banner campaigns for ∼2,000 participants
- **Ignite Designing & Drafting Team**: designed brochures, ID cards, posts and publicity for events and guests
- **Mentoring**: conducted ML workshops and tutored ∼300 students at BIT Mesra
- **Pratibimb Society**: guitarist, led stage shows, **1st prize** in band competition